

PRANAV SHANKAR

 github.com/PranavShankar  +1 (510)-565-4040  linkedin.com/PranavShankar  pranavs3@usc.edu

Education

University of Southern California, Los Angeles, CA

August 2024 – May 2026

Masters in Science - Computer Science

GPA: 4.0/4.0

Coursework: Analysis of Algorithms, Machine Learning, Database Systems, Natural Language Processing

PES University, Bangalore, India

June 2020 – June 2024

B.Tech. in Computer Science Engineering

GPA: 4.0/4.0

Coursework: Object Oriented System Design, Software Engineering, Machine Learning, Operating Systems

Technical Skills

Languages: Python, Java, C, C++, HTML, CSS, JavaScript, TypeScript, SQL, Go, Rust, Scala, R, Bash

Frameworks: ReactJS, NodeJS, NLTK, Bootstrap, Pandas, LangChain, Transformers, Pytorch, TensorFlow, Scikit-Learn

Databases and Tools: DynamoDB, Firebase, Git, Docker, Kafka, Kubernetes, Spark, Hadoop, Amazon SageMaker, Jira

Cloud and Knowledge Areas: AWS, GCP, Large Language Models, Software Development Life Cycle, Deep Learning

Experience

AMAZON WEB SERVICES (AWS) - AMAZON

May 2025 – August 2025 Certificate

Software Developer and Networking Intern

Seattle, WA

- Developed the **Blackfyre Capacity Simulator** to model **35k+ EC2 NX firmware tests** across **5,500+ EC2 instance droplets**, enabling precise capacity planning for infrastructure costing **\$18M/month**.
- Reduced execution runtime of EC2 Droplets from **32 to 25 hours** by resolving capacity bottlenecks using **Z3 SMT Solver**, driving **\$3M** in cost savings.
- Architected and deployed **scalable, serverless, and containerized** solutions leveraging AWS services including **DynamoDB, Lambda, EC2, S3, EFS, Fargate**, and the **Coral Framework**—significantly enhancing system performance and enabling the above operational efficiencies.

ARUP

January 2024 – June 2024 Certificate

Software Engineering Intern

Hyderabad, India

- Working with **LS-DYNA** in-house **Oasys pre- and post-processing finite element analysis software**
- Migrated and built **5+ automotive assessment templates** in **REPORTER** software, which consolidates results from the pre-processor **PRIMER** to post-processors **D3PLOT** and **T/HIS**
- Researched and developed **REPORTER** templates for critical automotive assessments including **GNCAP, USNCAP, UN Assessment**, and **KNCAP**, using **JavaScript, C, C++, and SVN**

Academic Projects and Publications

TrendTok: AI-Driven Music Synthesis and Trend Generation App | *GCP, Llama, MusicGen*

July 2024 Code

- Built web app replicating TikTok's UI using HTML, Bootstrap, and JavaScript, powered backend with Flask, and hosted on Google Cloud Platform, ensuring security and scalability through Docker.
- Utilized Selenium and BeautifulSoup for web scraping TikTok data, integrated third-party APIs, applied NLP techniques for text processing with Llama 3 API, generated music using Meta MusicGen, and created images with Stable Diffusion

Chatbot Application with AWS Bedrock | *Python, Docker, Langchain, Streamlit*

May 2024

- Performed analysis on the social media platform “Meet-Up” for Nashville City based on the data set.
- Utilized the principles of graph theory to analyze various **network analysis metrics, detecting cycles, finding geocentric and influencer nodes in the network.**

U.S Stock Market Portfolio | *MERN, AWS, Flask*

January 2023 Code

- Built a responsive, full-stack stock market analysis platform leveraging React.js, Bootstrap, and Node.js, enabling seamless integration of real-time stock data with MongoDB and deployed the application on AWS.
- Implemented RESTful APIs in the backend for real-time updates on price history, market trends, and user-specific portfolio tracking, enhancing usability and system reliability.
- Developed an Android application 'Stocks' in Java using the same Node.js backend ensuring cross-platform accessibility.

Toxic Comments Classification using Bidirectional LSTM | *Python, ML, LSTM* June 2023 Paper

- Developed a novel model for toxic comments classification which uses a combination of Bidirectional and Convolutional Layers. Achieved its accuracy of 99.6%, higher than pre-existing models
- Paper was presented in the IACIDS Conference and published at the prestigious EAI Index Journal**